

Georgios Pantazopoulos

+44 7873 489724 | georgepantaz94@gmail.com | [Website](#) | [Github](#) | [Scholar](#) | [LinkedIn](#)

Work Experience

AI Research Intern

Jun 2025 - Dec 2025

Alan Turing Institute

- Trained GUI agents with, supervised fine-tuning, chain-of-thought training, and reinforcement learning
- Developed data cleaning pipelines achieving 400% compression ratio with no performance degradation
- Trained ranker models to detect false-positive across 4.8M examples improving F1 by 13%

AI Engineer Intern

Feb 2025 - Apr 2025

Reka AI

- Developed contrastive models improving recall by 15% in image-to-image/video & text-to-video retrieval
- Contributed to search-augmented VLMs with embedding models for video retrieval
- Implemented instance retrieval from natural language descriptions in videos

Machine Learning Engineer

Sep 2018 - Apr 2020

Behavioral Signal Technologies Inc.

- Developed CNN-based classification models for emotion recognition from raw audio signals
- Built LSTM-based classification models for emotion recognition from automatic transcriptions
- Designed late/early-fusion methods for audio and text improving performance by 8% over unimodal models
- Led the data collection from crowdsourcing platforms resulting in the collection of 600k annotated examples

Research Experience

Retrievit

Dec 2025 - Mar 2026

University of Edinburgh, [\[code\]](#)

- Trained and tested Transformers, SSMs, and Hybrid interleaved/dual stream models for in-context retrieval
- Demonstrated that hybrids excel in length generalization while matching Transformers efficiency
- Conducted representational analysis in embedding spaces showing locality-preserving structures in SSMs/hybrids

VLMs as Visual Assistants

Dec 2024 - Jan 2025

Heriot-Watt University, [\[code\]](#)

- Showed that current VLMs are not successfully responding to Braille text from images
- Developed a dataset for Braille recognition by rendering text in images
- Trained models achieving more than 10-fold performance improvement in cross-script QA

Mamba-VL

Dec 2023 - Mar 2024

Heriot-Watt University, [\[code\]](#)

- Trained & Benchmarked 3 large-scale VLMs using Transformer and Mamba-based LLMs
- Benchmarked across 13 vision-language tasks spanning varying levels of perceptual granularity
- Demonstrated superior captioning, VQA, and reading-comprehension performance
- Used synthetic associative recall tasks to measure performance trade-offs

EMMA

Jan 2022 - May 2023

Alexa Prize Simbot Challenge, Finalist [\[code\]](#)

- Co-led the pre-training of the multimodal agent achieving state-of-the-art results in image understanding
- Fine-tuned the base model for hierarchical decision making enabling tool and boosting success rate by 7%
- Contributed to self-play synthetic data generation pipelines doubling task success

Exploring the Role of Patch Resolution for Vision Transformers

Jan 2022 - Mar 2022

University of Edinburgh

- Pretrained vision transformers from scratch
- Demonstrated that ViTs have sharp attention score distributions in deeper layers
- Replaced self with linear attention achieving $2.3 \times$ inference speed with minimal performance loss

Education

PhD in Robotics and Autonomous Systems

Sep 2021 - July 2026

CDT in Robotics and Autonomous Systems, University of Edinburgh, Heriot-Watt University

- Thesis: From Pixels to Actions: Grounding in Vision-Language Models, supervised by [A. Suglia](#)

MSc in Artificial Intelligence with Speech and Multimodal Interaction

Sep 2020 - Aug 2021

Heriot-Watt University, (1st with Distinction)

- Thesis: Compositional Generalization in Image Captioning, supervised by [A. Eshghi](#)

MEng in Electrical and Computer Engineering
National Technical University of Athens

Sep 2012 - Jun 2018

- Thesis: Stress Detection from Speech Signals, supervised by [A. Potamianos](#)

Selected Publications

Retrievit: In-context Retrieval Capabilities of Transformers, State Space Models, and Hybrid Architectures
2026 *Transactions on Machine Learning Research*, [[paper](#)]

- [G.Pantazopoulos](#), M.Nikandrou, I.Konstas, A.Suglia

An Efficient Training Pipeline for Reasoning Graphical User Interface Agents
2026 *ICLR 2026 Workshop on Multimodal Intelligence*, [[paper](#)]

- [G.Pantazopoulos](#), E.Özyiğit

Evaluating Multimodal Language Models as Visual Assistants for Visually Impaired Users
2025 *Annual Meeting of the Association for Computational Linguistics*, [[paper](#)]

- [A.Karamolegkou*](#), [M.Nikandrou*](#), [G.Pantazopoulos*](#), [D.Sanchez](#), [P.Rust](#), [R.Dhar](#), [D.Hershcovich](#), [A.Søgaard](#)

CROPE: Evaluating In-Context Adaptation of Vision and Language Models to Culture-Specific Concepts
2025 *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, [[paper](#)]

- [M. Nikandrou](#), [G. Pantazopoulos](#), [N. Vitsakis](#), [I. Konstas](#), [A. Suglia](#)

Shaking Up VLMs: Comparing Transformers and Structured State Space Models for Vision & Language Modeling
2024 *Conference on Empirical Methods in Natural Language Processing*, [[paper](#)]

- [G. Pantazopoulos](#), [M.Nikandrou](#), [A. Suglia](#), [O. Lemon](#), [A. Eshghi](#)

Enhancing Continual Learning in Visual Question Answering with Modality-Aware Feature Distillation
2024 *ACL Third Workshop on Advances in Language and Vision Research*, [[paper](#)]

- [M.Nikandrou*](#), [G. Pantazopoulos*](#), [I. Konstas](#), [A. Suglia](#)

Lost in Space: Probing Fine-grained Spatial Understanding in Vision and Language Resamplers
2024 *Annual Conference of the North American Chapter of the Association for Computational Linguistics*, [[paper](#)]

- [G. Pantazopoulos](#), [A. Suglia](#), [O. Lemon](#), [A. Eshghi](#)

Multitask Multimodal Prompted Training for Interactive Embodied Task Completion
2023 *Conference on Empirical Methods in Natural Language Processing*, [[paper](#)]

- [G. Pantazopoulos](#), [M.Nikandrou](#), [A. Parekh](#), [B. Hemanthage](#), [A. Eshghi](#), [I. Konstas](#), [V. Rieser](#), [O. Lemon](#), [A. Suglia](#)

Data Augmentation using GANs for Speech Emotion Recognition
2019 *Annual Conference of the International Speech Communication Association*, [[paper](#)]

- [A. Chatziagapi](#), [G. Paraskevopoulos](#), [D. Sgouropoulos](#), [G. Pantazopoulos](#), [M.Nikandrou](#), [T. Giannakopoulos](#), [A. Katsamanis](#), [A. Potamianos](#), [S. Narayanan](#)

Teaching Experience

AθNLP 2024 / 2025

Sep 2024 & Sep 2025

Summer School Teaching Assistant, [[code](#)]

- Developed the lab materials for sequence modeling and multimodal models
- Delivered the material in lab sessions targeting over 100 students with varying prior experience

Introduction to Natural Language Processing & Data Mining and Machine Learning

Sep 2022 - Dec 2025

Teaching Assistant, [[code](#)]

- Co-led the infrastructure and the coursework in NLP and ML courses
- Supervised groups of postgraduate students on their semester project

Skills

Programming Languages

Python, Matlab, R, C, C++, C#, Java, PDDL, Bash, SML

ML Frameworks

PyTorch, Hugging Face, Transformers, PEFT, Accelerate, TRL,
PyTorch Lightning, scikit-learn, spacy, openCV, Albumentations, YOLO, Gradio

Data Engineering

Pydantic, Pandas, PyArrow

MLOPs & CI/CD

Weights & Biases, MLflow, Github Actions

Vector Databases

Faiss, Milvus, chroma, Weaviate

Training & Inference

Slurm, Docker, Kubernetes, vLLM

Evaluation & Benchmarking

large-scale data collection & filtering, data and model ablations